

Truth-Tracking Profiles: What LLMs Participate In

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Abstract

Debates about large language model grounding often ask whether model representations connect to the world. This paper argues that the framing is too coarse. The representational successes picked out by *true* are supported by a projectible profile of partially coupled stabilisers: perception, testimony, coherence, measurement, intervention, error-correction, and practical success. The profile does not define truth and need not be homeostatic throughout. It explains why truth-relevant successes travel together, why they fail in patterned ways, and what can be predicted from a system’s stabilising background.

Language models are the pressure case. Text-trained systems inherit dense textual traces of testimony and local coherence while weakly participating in perception, measurement, intervention, and practical correction. The result is a predicted failure signature: fluent, coherent outputs that lose answerability when tasks require independent routes to the facts. Retrieval, tools, multimodal input, and feedback change the profile, so the question is comparative: which stabilisers does a system participate in, how tightly are they coupled, and what failures should that pattern lead us to expect?

I INTRODUCTION: A BINARY PRESUPPOSITION

Do the representations inside a large language model connect to the world? The question is usually put as a yes/no. Bender and Koller (2020) say no: a system trained on form alone, like an octopus eavesdropping on telegraph traffic, learns the statistics of language without touching what language is about. Others reply that such systems meet at least some conditions on reference, or that sensory grounding isn’t required for thought in the first place (Chalmers, 2023; Coelho Mollo & Millière, 2026).

The camps disagree on the verdict but share the question’s shape.

That yes/no framing makes world-connection look like a single capacity, something a system either has or lacks. The successes we call truth-tracking are instead produced by distinct stabilising mechanisms that ordinarily operate together: perceptual contact, testimonial inheritance, inferential

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coherence, measurement and intervention, error-correction, and practical success. A system can take part in some of these while sitting out the rest. Text-only and weakly tooled language models make that possibility especially visible.

The paper’s claim is that those stabilisers form a projectible profile around the representational successes picked out by the predicate *true*: a pattern whose point is prediction and explanation rather than classification for its own sake (Goodman, 1955; Khalidi, 2013). LLMs matter because they make the dissociation visible: they inherit dense textual traces of testimony and local coherence patterns while lacking much of the perceptual and verificational coupling that normally keeps claims answerable to the world.

The claim is clearest for systems whose contact with the world remains mediated mainly by human text. Multimodal models, retrieval systems, tool-using agents, and systems embedded in experimental feedback loops change the profile. They may add some of the missing stabilisers, strengthen others, or introduce new failure modes. A profile view should track those differences rather than defend a blanket verdict about all LLMs.

The contribution is diagnostic. Boyd (2021) treats reference and truth as special cases of accommodation: the alignment of perceptual, instrumental, cognitive, and representational practices with causally relevant structures. This paper asks what happens when a system enters some of that alignment through human text while missing other routes through which accommodation is normally maintained. LLMs inherit textual products of testimony and coherence, while lacking or weakening perception, measurement, intervention, and world-directed correction. The profile view predicts that this partial participation should have a characteristic failure signature.

That prediction is a hypothesis about a profile. It should move with the evidence. If a system gains reliable routes through perception, measurement, action, or external checking, the predicted vulnerability should weaken or change shape. If it doesn’t, the profile account has misdescribed the stabilisers.

That diagnostic has to be earned, because the profile view invites an obvious complaint. The component theories each isolate a genuine stabiliser. Reliabilism gets process reliability right (Goldman, 1979); coherentism gets inferential integration and the demand for truth-conducive input right (BonJour, 1985); pragmatism gets the corrective role of action and success right; social epistemology gets testimonial and institutional transmission right (Coady, 1994; Goldberg, 2010; Lackey, 2008).

The relabeling worry is that profile talk only bundles these local insights and calls the bundle “projectible”. That worry is fair. Profile talk adds something only where coupling among stabilisers supports predictions that source-by-source assessment leaves out.

The paper develops three such places. First and most sharply, the profile structure predicts a characteristic failure signature: pull the stabilisers apart, and the account says in advance what should go wrong. Second, it makes conflicts between mechanisms more tractable, by treating their degree of co-variation as evidence in its own right. Third, it predicts where truth-relevant mechanisms should co-vary tightly across domains and where they should come apart.

The language-model case is where the first prediction pays off. A text-only system trained on human writing is strongly exposed to the textual traces of testimony and to local patterns of coherence, while lacking the standing of a normal speaker or hearer in a testimonial exchange. It also lacks direct perceptual contact, independent measurement, and ordinary correction through action in the world. The profile view therefore predicts a distinctive vulnerability for such systems: fluent and internally

coherent outputs can track textual regularities while failing to remain answerable to the facts those regularities were originally used to report.

Human beings hallucinate, misremember, and confabulate too. The contrast lies in the profile of correction. In ordinary human cases, those failures occur against a background of perception, action, and social correction; in a text-only model, several of those world-corrective paths are absent or attenuated as part of the standing profile. What the LLM literature calls hallucination is one expression of that vulnerability (Huang et al., 2025).

As models acquire external tools, retrieval, multimodal inputs, or action-guided feedback, this contrast should be re-estimated rather than repeated unchanged. The argument's object is the pattern of participation rather than the label *LLM*.

That vulnerability marks the distance between the formal linguistic competence these models plainly have and the functional competence that world-contact underwrites (Mahowald et al., 2024).

The argument runs as follows. Section 2 fixes the target: the relevant cases are representational successes (beliefs, utterances, measurements, models, inferential outputs), not truth as an abstract object. Section 3 sets out the stabilising mechanisms and their Boydian lineage.

Section 4 says how the profile supports projection, locating projectibility in inferential, practical, and corrective success under counterfactual perturbation rather than in survival or usefulness alone. Section 5 makes good on the conflict and co-variation claims, the core of the reply to relabeling. Section 6 takes objections. Section 7 develops the language-model profile mechanism by mechanism. Section 8 concludes.

2 THE TARGET: TRUTH-RELEVANT REPRESENTATIONAL SUCCESS

The target has to be fixed before the profile can do any work. Truth, considered as an abstract property or as the role of the truth predicate, is not the sort of thing that has perceptual, testimonial, or instrumental mechanisms as parts. The target cases are representational successes: beliefs, assertions, inscriptions, measurements, models, and inferential outputs assessed as getting things right.

Some of these are central TRUTHBEARERS. Beliefs and assertions can be true or false in the ordinary sense. Others are neighbouring cases that support truth-ascription by yielding, recording, or constraining claims: a measurement, a map, a statistical model, or an experimental display may be accurate, calibrated, or well-fitted in ways that make some associated propositions true. The profile concerns that broader space of truth-relevant representational success.

This already narrows the ambition. The paper leaves a reductive definition of truth to one side. Deflationary and minimalist theories can insist that the truth predicate has a thin logical or generalising role, and that the basic theory of truth shouldn't be loaded with a substantive causal story (Horwich, 1998). The profile account can grant that point. It asks why some truth-relevant representational successes travel together, support projection, and fail in patterned ways.

The same separation keeps the correspondence intuition in place without making correspondence the whole theory. Correspondence theorists are right that truth involves answerability to how things are (David, 1994; Kirkham, 1992).

For present purposes, ANSWERABILITY names a constraint relation, not a new theory of truth. A representation is answerable to a target when failures in how the target is can bear on the production, uptake, correction, or withdrawal of the representation. A calibrated instrument has answerability because relevant discrepancies can show up in checks, recalibrations, and downstream failures. A

coherent but disconnected generator lacks it where no route lets the relevant fact constrain the output. The profile describes how such routes combine and come apart.

Pragmatist theories mark a different stabiliser. Inquiry, action, and successful practice can correct representations that are locally coherent or socially inherited but poorly connected to the world (Dewey, 1941; James, 1907). That doesn't make truth identical with what works. It says that practical success is one way representational systems become answerable to what they are about.

The explanandum is the stabilising background that makes truth-ascription, accuracy-ascription, and truth-relevant representational success projectible across cases. We ask why some outputs fall inside the extension, why nearby outputs fall outside it, and which features let us project from a partial description of a case to its likely epistemic behaviour. This is where Goodman's projectibility problem enters the paper: the issue is which patterns make truth-relevant classifications reliable guides to further expectation, rather than how to attach the word *true* case by case.

Boyd's accommodationism supplies the background for that move. Reference and truth are not secured by definitions in isolation; they are achievements of practices that align perceptual, instrumental, cognitive, and representational activity with relevant causal structures (Boyd, 2021). The present paper accepts that frame and asks a narrower question: how strongly does a given system participate in the practices and mechanisms that make such alignment reliable?

For language models, the lesson is simple. Their outputs are assessed through some stabilisers and not others. A text-only model may inherit textual traces of testimony, statistical regularities, and local coherence pressures while lacking perception, measurement, intervention, and ordinary practical correction. That profile is enough to make truth-relevant assessment possible, uneven, and empirically diagnostic.

3 TRUTH-TRACKING AS A PROJECTIBLE MECHANISM PROFILE

Goodman's new riddle fixes the projectibility problem: the hard question isn't whether past cases support some generalisation, but which predicates support projection from known to unknown cases (Goodman, 1955). The present paper asks that question about truth-relevant representational successes: which stabiliser patterns let us project from partial evidence to likely success or failure?

Khalidi's account supplies the network discipline behind this profile talk. Natural categories earn their status by supporting prediction and explanation, and they do so when associated properties stand in causal relations that make some properties projectible from others (Khalidi, 2013).

The present use is an adaptation. The target cases are the representational successes fixed in Section 2; the stabilisers explain why those cases support projection. Perception, testimony, measurement, coherence, intervention, correction, and practical success are linked routes through which representational outputs become more or less answerable to the world. Khalidi's later "nodes in causal networks" formulation gives the graph vocabulary for this representation (Khalidi, 2018).

This profile view is Boydian in lineage. Homeostatic property cluster (HPC) theory explains how real kinds can be projectible without essences (Boyd, 1991, 1999). Boyd's later accommodationism supplies the semantic background: successful reference is an achievement of whole practices, not a relation fixed term by term by accepted definitions. Perceptual skills, instruments, inferential routines, social arrangements, and representational uses all contribute when they help a practice align with relevant causal structures (Boyd, 2021).

The present claim is narrower than Boyd's. It asks how the accommodation profile changes when

the participant is a text-trained system. An LLM can inherit stabilized linguistic and testimonial patterns while missing the instrumental, perceptual, and practical routes that normally keep those patterns answerable. Stable or projectible profiles shouldn't automatically be called homeostatic; that stronger description should be reserved for cases where corrective coupling has been shown (Reynolds, 2026).

That means the paper is continuous with accommodationism at the explanatory level and diagnostic at the level of system profile. Accommodationism tells us why representational practices can be answerable to causal structure; the profile account tracks how strongly a given system participates in the stabilisers that make that answerability possible. LLMs are the pressure case because they inherit accommodation through human text while missing or weakening several world-corrective paths.

Boyd's homeostatic property cluster is the important special case in which links are actively maintained, where mechanisms correct one another so that the properties co-occur far more reliably than chance. The truth-tracking profile has tightly coupled regions, and the connections thin toward its edges. The profile view doesn't need the whole field to be literally homeostatic before it can project.

That picture does two jobs. The tightly coupled core, everyday perception and basic coordination, is where mutual correction is strongest, and the coupling there is what predicts a language model's failure signature: remove the perceptual and verificational nodes, and representational success should fail in a specific way (Section 7).

It places the loose cases. Toward the edges, in theoretical physics, ethics, and some formal domains, the nodes are real but sparsely linked. Reading the profile as a network doesn't by itself fix where its boundaries fall: Onishi and Serpico (2022) show that carving a kind from a causal graph stays relative to the chosen level of abstraction, the interest-relativity Craver (2009) found in mechanism talk.

So the paper fixes the epistemic purpose. Following Khalidi's distinction between epistemic purposes and non-epistemic practical or aesthetic ones, the profile is individuated by representing a target domain well enough for relevant inferential, practical, or corrective success under perturbation (Khalidi, 2013). Relative to that purpose, its connections can thin at the edges without the profile dissolving.

4 WHY THE PROFILE IS PROJECTIBLE

Section 2 fixed the target cases, and Section 3 identified the stabilisers. Projectibility concerns the relation between them. The question is whether a partial description of an output and its stabilising background lets us predict further truth-relevant success or failure.

The guiding purpose is representing a target domain well enough for relevant inferential, practical, or corrective success under perturbation. Perturbation matters because it distinguishes lucky fit from robust answerability. A representation may succeed in the training cases, the familiar environment, or the current conversational context. It projects more strongly when it continues to guide action, inference, measurement, and correction after conditions change.

Survival and reproduction are limiting biological cases of this structure, not the general explanation. Hoffman-style interface arguments warn that adaptive success and veridical representation can come apart (Hoffman et al., 2015). Martínez (2019) argues that usefulness can drive representations toward truth under specified conditions. The lesson for this paper is methodological: usefulness

isn't itself the truth-making relation. It becomes truth-relevant only when the task, environment, and correction channels make error costly in ways that expose the relevant mismatch.

The same point holds outside biology. In science, law, medicine, engineering, testimony, and model evaluation, the stabilisers are often institutional and technological. Replications, instruments, standards, records, peer challenge, and downstream practical failure belong to truth-relevant assessment. They are among the routes by which a truth-relevant output becomes robust under perturbation.

The profile supports projection when such routes are coupled. A perceptual judgement supported only by immediate appearance projects weakly if it is easily overturned by lighting, perspective, testimony, or measurement. The same judgement projects more strongly when it is embedded in a practice where perception, action, memory, testimony, and correction have repeatedly calibrated one another.

This is why the privileged stabiliser can change across domains. Some mathematical and logical cases project through inferential structure with little perceptual input. Some scientific cases project through instrumentation and replication rather than ordinary observation. Some testimonial cases project because the institution has mature roles, records, and correction channels. The projectible unit is the coupled pattern.

This also explains why edge cases are expected. Where stabilisers thin or fail to calibrate one another, truth-relevant classifications should project less strongly. Frontier science, remote history, contested legal facts, and ethical disagreement should be treated as thinner regions of the profile. They may still involve real stabilisers, but the pattern supports weaker expectations about agreement, correction, and future success.

The language-model prediction follows directly. A system that strongly inherits textual traces of testimony and local coherence, while weakly participating in perception, measurement, intervention, and practical correction, should show a specific failure profile. It can produce fluent outputs that look truth-relevant from inside the textual profile while failing under perturbations that require independent contact with the world.

Tool use, retrieval, multimodal input, expert review, and action-guided feedback matter because they change the stabiliser pattern. They may strengthen projectibility in some domains, leave it weak in others, or introduce new routes to error. The profile view therefore predicts comparative differences among systems and tasks, rather than a single verdict about whether language models are grounded or ungrounded.

5 WHAT THE PROFILE STRUCTURE DOES THAT COMPONENT THEORIES DON'T

This section makes good on the claim from Section 1 that the profile view earns its keep. The useful test is whether profile structure answers questions that are hard to formulate when one stabiliser is treated as the main unit of assessment. The answer turns on coupling: the evidential fact is sometimes the way mechanisms have calibrated one another, rather than the bare performance of any one mechanism taken in isolation.

The component theories enter as local successes. Goldman-style reliabilism rightly asks about belief-forming processes and their conditional reliability (Goldman, 1979). BonJour-style coherentism rightly treats inferential and explanatory integration as epistemically significant, while also insisting that coherence needs input if it is to be truth-conducive (BonJour, 1985). Pragmatist accounts

rightly stress correction through successful practice (Dewey, 1941; James, 1907). Correspondence accounts rightly preserve answerability to the world (David, 1994; Kirkham, 1992).

Social epistemology rightly treats testimony and institutions as genuine epistemic sources (Coady, 1994; Goldberg, 2010; Lackey, 2008).

The profile view begins after those successes are granted. Its unit is the pattern of calibration among stabilisers: perception checked by testimony, testimony disciplined by memory and records, coherence constrained by measurement, measurement corrected by intervention, and practice pressured by failure. Read as a causal network (Section 3), the profile has structure that source-by-source assessment leaves in the background: the strength and history of the links between nodes.

5.1 ADJUDICATING MECHANISM CONFLICT

When two stabilisers disagree, say perception reports one thing and well-attested testimony another, a theory focused on one source at a time underdescribes the evidential situation. Goldman-style reliabilism can ask which process type is reliable, including conditional reliability; social reliabilism can extend the assessed process through testimonial production (Goldberg, 2010; Goldman, 1979). Those are genuine resources. The profile question is different: how tightly have these sources been calibrated against one another in this domain?

In Khalidi's terms, causal relations among associated properties underwrite projection; here the relevant projection runs from one stabiliser's output to another's (Khalidi, 2013).

Consider a modest conflict. A subject's visual judgement says that an object has property *P*. A calibrated instrument says that it lacks *P*. Several competent people, drawing on the same instrument tradition, say the instrument is functioning normally. The profile view avoids a standing rule to trust perception, testimony, or the instrument. It asks which sources have historically corrected which others in this region of practice.

In everyday object recognition, ordinary perception, practical feedback, memory, and testimony have long corrected one another. If I seem to see an old friend at the end of a hallway while well-placed testimony rules out her presence, the perceptual judgement can reasonably give way. Coady's discussion of perception, memory, inference, and testimony already pushes in this direction: the sources interpenetrate, and testimony can rationally undermine what presents itself as perception (Coady, 1994).

In a mature instrumental domain, the priority can reverse. Unaided perception may be the weak node, because the tighter profile joins calibrated instruments, records, expert testimony, replication, and repeatable intervention. The same abstract conflict therefore receives different treatment in different domains. Global source reliability is too coarse a contrast. The history of cross-source correction gives one output a stronger projective relation to the rest of the profile.

This makes co-variation epistemically load-bearing. A tightly coupled profile gives a defeasible presumption that a lone divergent output is local error or a boundary case. A loosely coupled profile gives a different presumption: divergence is evidence that the kind of success under assessment projects weakly here. The verdict is still empirical and fallible, but the evidence includes the pattern of calibration among stabilisers, rather than the assessed reliability of one source alone.

That difference matters when two descriptions agree about local reliability but disagree about transfer. Suppose a model, instrument, or witness has been reliable in a domain where perception, records, and measurement have repeatedly corrected one another. A source-by-source assessment can preserve that reliability score when the source is moved to a neighbouring domain.

The profile view doesn't automatically preserve it. If the new domain lacks the calibration history that made the earlier success projectible, it predicts weaker transfer even before any present disagreement appears. That is the divergent prediction: reliability inherited without its cross-source correction background should degrade under perturbations that remove the missing checks.

5.2 DOMAIN-CLUSTERED CO-VARIATION

The network view also foregrounds a cross-domain pattern that source-by-source accounts tend to leave implicit: disagreement should cluster where the independent marks of connectivity are thin. Khalidi treats causal strictness and network density as dimensions along which categories support stronger or weaker projection (Khalidi, 2013). The stabilisers of truth-relevant success should be densely linked where they have been jointly disciplined, and sparsely linked where they have not. Persistent disagreement should therefore cluster in the sparsely connected regions.

The prediction is non-circular only if connectivity is characterised without appeal to whether the mechanisms agree, and it can't be read off the graph in a purpose-free way: Onishi and Serpico (2022) show the abstraction level and boundaries stay interest-relative.

The reply is to fix the purpose rather than disown it. Holding the representational aim of Section 4 fixed, connectivity can be characterised by three marks established before present agreement is checked.

First, the stabilisers can share a selection or training history. Perception, naming, action, and memory in ordinary object recognition have been shaped by recurring tasks and recurring error signals. Instruments, calibration routines, and expert reports in mature scientific practices can share a similar history, even when the relevant objects are not available to unaided perception.

Second, the practice can have institutional age. Age by itself is not enough; a long-lived mistake remains a mistake. The relevant sign is the accumulation of correction norms, records, role differentiation, and procedures for passing error reports forward. These features make one stabiliser available as a check on another before any present disagreement has been adjudicated.

Third, the domain can have measurement infrastructure. Instruments, standards, replications, and intervention routines let a practice ask whether perception, testimony, or coherence has drifted. Where those infrastructures are thick, the stabilisers can correct one another. Where they are thin, the same stabilisers may continue to operate, but their relations support weaker projection.

These marks are defeasible indicators, not criteria. Disagreement can also be driven by stakes, incentives, ideology, small samples, strategic ambiguity, or value-ladenness. The profile view doesn't predict disagreement from thin connectivity alone. It predicts that disagreement will be especially persistent where those familiar pressures combine with weak cross-source correction.

Once the purpose is fixed, the profile therefore predicts tighter agreement where shared training history, institutional correction, and measurement infrastructure are thick. It predicts looser agreement where those marks are absent. That is why frontier science, ethics, contested law, and remote history are expected edge cases: some stabilisers remain real while the links among them thin.

This is the sense in which profile structure adds to the component theories. Process reliability, coherence, correspondence, pragmatic success, and testimony remain local successes. The profile explains why their relative epistemic force changes across domains and why their separation has a characteristic signature. Section 7 applies that result to language models: systems can inherit textual and coherentist stabilisers while lacking the perceptual and verificational links that usually keep those stabilisers coupled.

6 OBJECTIONS

6.1 INTEREST-RELATIVITY

One objection cuts at the profile structure itself. Onishi and Serpico (2022) argue that Khalidi's causal-network account inherits the interest-relativity Craver (2009) finds in mechanistic accounts: which causal graph you draw, at what level of abstraction, and where you set its boundaries all answer to explanatory interest, so the network never fixes a unique, interest-free carving of kinds. If that's right, the profile of truth-relevant success is only as objective as our purposes.

The reply concedes the premise and limits the consequence. The account makes two claims, both purpose-relative and both revisable: systems missing perceptual and verificational stabilisers should show the failure signature of Section 7, and connectivity should track selection history, institutional age, and measurement infrastructure (Section 5). Each holds once the representational purpose is fixed, and either can fail if the stabilisers are misdescribed.

Onishi and Serpico (2022) reach the same general verdict, that interest-relativity is ineliminable, and Khalidi's domain pluralism leaves room for this move (Khalidi, 2013). This account makes the operative interest explicit instead of disowning it.

6.2 RELABELING AND ABSORPTION

A second objection says that the account will be absorbed by the component theories. Reliabilism can already conditionalise process reliability. Social epistemology can extend the assessed process through testimony, records, and institutions. Coherentism can demand input constraints. On this objection, the profile view redescribes familiar epistemology in causal-network language.

The reply is not that the component theories lack the resources to mention coupling. They often have them. The claim is comparative: profile structure makes the calibration history itself the unit from which we project. Section 5 gave the central case. A source can retain its local success record while losing the correction background that made that success travel. The profile view predicts weaker transfer in that case, because the missing cross-source links matter before a fresh reliability score has been established.

That is why the account shouldn't be read as a replacement for reliabilism, coherentism, correspondence, pragmatism, or social epistemology. It is a way of saying when their local stabilisers support projection together, when they pull apart, and what failures should follow.

6.3 DERIVATIVE TESTIMONY

A third objection concerns testimony. Testimony is not just information with a human source. It is an interpersonal epistemic practice involving assertion, responsibility, trust, uptake, and defeaters (Coady, 1994; Goldberg, 2010; Lackey, 2008). If so, it is misleading to give a language model testimonial standing merely because it was trained on human text.

The objection is right, and the account should use the weaker claim. A text-trained model inherits artifacts of testimonial practices: reports, explanations, records, instructions, arguments, and corrections left in text. That inheritance is a genuine stabiliser because many such texts were produced inside practices already partly accommodated to the world. It is also derivative. The model is not, by training alone, a normal hearer who can trust a speaker under standing norms of assertion, nor a speaker who takes responsibility for what is asserted. That derivative status explains both the reach and the fragility of text-only systems.

6.4 NON-EMPIRICAL DOMAINS

A fourth objection comes from mathematics, logic, ethics, fiction, and law. In these domains, perception, measurement, intervention, and practical success seem peripheral or misplaced. If the profile is built around world-contact, it may overfit ordinary empirical truth.

The answer is that no single stabiliser is mandatory across the whole profile. In mathematical and logical cases, inferential structure, proof norms, notation, memory, testimony, and error-correction carry much of the projection. In legal and ethical domains, institutional procedures, role differentiation, public reasons, precedents, and contestation can be stabilising even when agreement remains fragile. Fictional truth has its own textual and practice-bound constraints. These are thinner or differently coupled regions, not counterexamples to the profile structure.

6.5 DEFLATIONISM AND PLURALISM

Finally, deflationists and alethic pluralists may object from opposite directions. Deflationists may say that truth has no deep nature for the profile to explain (Horwich, 1998). Pluralists may say that different domains involve different ways of being true, so a single profile is artificial.

Both objections help fix the ambition. The paper isn't analysing the truth predicate or reducing truth to stabilisers. It is explaining projectible patterns among truth-relevant representational successes. That target is compatible with a thin account of the truth predicate. It is also compatible with domain pluralism, because the profile already allows different stabilisers to be privileged in different regions. The claim is not that truth is one homeostatic kind. It is that truth-relevant success often has a projectible stabiliser profile.

7 THE LLM PAYOFF: PARTICIPATION, NOT A VERDICT

The language-model payoff is not a verdict about a class. It is a way of fixing the bearer and projecting a failure profile. A pretrained text model, an instruction-tuned chat model, a retrieval-augmented system, a tool-using agent, a multimodal model, and an experimentally embedded robot do not have the same stabilising background. On the present view, that isn't a complication added after the theory. It is the theory's main prediction: change the stabilisers and the expected successes and failures should change too.

The key term is **PARTICIPATION**. A system participates in a stabiliser to the extent that its outputs are generated, constrained, corrected, or selected by that stabiliser in a way that supports projection to further cases. Participation can be inherited, as when training text contains the residue of human perception, testimony, measurement, and correction. It can be operational, as when retrieval, tools, sensors, or instruments constrain the present output. It can be corrective, as when feedback changes later behaviour. It can also be merely decorative, as when an output mentions experiments, sources, or measurements without being constrained by any of them.

The baseline text-only case is the cleanest pressure test. Such a model inherits textual products of human accommodation: reports written by perceivers, explanations corrected by teachers, records produced by institutions, descriptions shaped by instruments, and genres whose norms often reward coherence and accuracy. That inheritance is why text-only systems can do real epistemic work. It is also why the inherited stabiliser shouldn't be called ordinary testimony without qualification. The model has access to testimonial residue, not the standing of a speaker or hearer in a testimonial exchange.

System profile	Added or dominant stabiliser	Directional prediction
Text-only pretrained model	Textual inheritance and local coherence	Strong where dense, stable text is a good proxy; fragile for recent, obscure, perceptual, or measurement-dependent facts.
Instruction-tuned or RLHF model	Preference, helpfulness, safety, and benchmark feedback	Better conversational fit and some error reduction; no automatic independent route to the facts.
Retrieval-augmented system	Live access to records and institutional traces	Better on updated or source-backed recall; still weak where the missing route is experiment, perception, or intervention.
Tool-using system	Calculators, databases, code, instruments, or APIs	Better where the right tool is called and interpreted; new failures from tool choice, stale data, and integration errors.
Multimodal system	Perceptual input in image, audio, video, or sensor form	Better for some perceptually anchored tasks; not automatically calibrated measurement or practical correction.
Action-guided or expert-reviewed system	Practical feedback, review, and downstream failure pressure	Stronger correction profile where feedback is timely and domain-competent; weaker where review is sparse or proxy-based.

Table 1: Architecture-sensitive profile predictions.

Coherence is the second strong stabiliser. Training and decoding reward local well-formedness, genre fit, inferential smoothness, and continuation from context. Those pressures explain why these systems can display impressive formal linguistic competence, in the sense separated from functional world use by Mahowald et al. (2024). They also explain the vulnerability. Coherence can improve the shape of an answer without adding an independent route by which the relevant facts constrain it.

Perception, measurement, intervention, and practical correction are weak in the baseline profile. They enter indirectly through text produced by others, not as live routes by which the current output is checked. A model can generate a plausible lab result, legal summary, local observation, or bibliographic claim because similar text exists. That doesn't mean that an experiment, court record, observation, or catalogue entry has constrained the present output. The profile view therefore predicts failures exactly where textual inheritance and coherence cease to be good proxies for the target domain.

This gives hallucination a more specific role. It is not a single defect with one cause, and the LLM literature rightly treats it as heterogeneous (Huang et al., 2025). The profile claim is narrower: text-only systems should be especially vulnerable to fluent, coherent, genre-appropriate outputs that fail when the task requires a fresh or independent route to the facts. Those failures should be common for obscure sources, recent events, local observations, novel measurements, fine-grained quantities, and action-sensitive causal claims. They should be less common where dense, stable, redundant text

is already a good proxy.

This prediction is different from the generic claim that language models hallucinate. It says where mitigation should and shouldn't help. Self-consistency and related prompting strategies should improve coherence-sensitive performance more than independent answerability. Retrieval should help when the missing stabiliser is access to updated records or institutional testimony; it should help less when success depends on new perception, experiment, or intervention. This is why retrieval can reduce hallucination in conversation (Shuster et al., 2021) without solving the broader profile problem.

Tool use changes the profile again. A calculator, database, theorem prover, code interpreter, search system, or scientific instrument can add operational participation in measurement or record access. But the stabiliser is present only where the tool is actually called, appropriate to the task, correctly interpreted, and integrated into the answer. Otherwise tool talk is decorative. The failure profile shifts from pure fabrication toward tool-selection errors, stale sources, over-trust in bad outputs, and misinterpretation of results.

Multimodality also changes the profile unevenly. Visual, auditory, or sensor input can give a model operational participation in perception. That is a real difference from text-only inheritance. But perceptual input is not automatically calibrated measurement, intervention, or action-guided correction. A multimodal model may improve on image-bound tasks while remaining weak on tasks that require measurement standards, experimental control, or practical feedback.

Instruction tuning, RLHF, benchmark evaluation, red-teaming, and expert review are correction channels, but they must be classified by what they correct. They often impose helpfulness, harmlessness, preference satisfaction, conversational fit, and benchmark success. Those constraints can reduce some errors, but they can also reward answer-shaped behaviour when users or raters prefer confidence, agreement, or plausible explanation. Work on sycophancy shows the risk of treating preference feedback as truth feedback without further structure (Shapira et al., 2026). In the profile terms, these processes strengthen some social and evaluative stabilisers while leaving perception, measurement, and intervention proxy-based.

This is where Coelho Mollo and Millièrè are the live interlocutors, not a foil. Their vector-grounding argument distinguishes kinds of grounding and argues that some LLM states can satisfy causal-historical or teleosemantic conditions for referential grounding (Coelho Mollo & Millièrè, 2026). The present account needn't deny that. Its explanandum is broader than referential grounding. Referential grounding is one dimension of world-connection; truth-tracking answerability depends on a wider profile of stabilisers, including correction, measurement, intervention, perception, and practical feedback. An LLM state might be referentially grounded in their sense while the deployed system still has a fragile truth-tracking profile for tasks requiring independent checking.

The upshot is a set of empirical expectations. Reliability should vary not simply with topic difficulty, but with the stabiliser structure of the task. Text-only systems should be strongest where textual inheritance and local coherence are good proxies for truth-relevant success. Retrieval should shift the profile mainly for record access. Tools should shift it where measurement or calculation is actually routed through the tool. Multimodality should shift perceptual tasks without automatically solving intervention or calibration. Action-guided agents should improve where downstream failure provides fast, task-relevant correction.

These expectations are defeasible. If reliable perception, measurement, retrieval, tool use, expert correction, or action feedback fails to alter the predicted failure signature, then the profile has mis-

described the mechanisms. If a text-only system remains mainly dependent on inherited textual accommodation and local coherence, the same vulnerability should persist even as fluency improves. That is the point of treating participation as a profile: it lets the grounding question become a prospective diagnosis rather than a classificatory verdict.

8 CONCLUSION

The question that opened the paper was whether the representations inside a large language model connect to the world. The answer has been a change in what the question asks. World-connection is better treated as a profile of participation in the mechanisms by which representational outputs become answerable to what they are about.

The central claim has been that truth-relevant representational success is supported by a projectible profile of stabilisers: perception, testimony, coherence, measurement, intervention, error-correction, and practical success. These stabilisers do not define truth, and the account does not compete with thin accounts of the truth predicate as a semantic analysis. Its target is different: it explains why the cases picked out by successful truth-ascription travel together, fail in patterned ways, and support predictions from partial knowledge of a system's stabilising background.

That profile has a tightly coupled core and thinner edges. Where perception, action, memory, testimony, instruments, and practical failure have repeatedly corrected one another, the coupling can become homeostatic in Boyd's strong sense. Toward frontier science, ethics, formal domains, law, and remote history, the nodes stay real while the links thin. The profile can remain projectible there without being homeostatic throughout, and it supports correspondingly weaker expectations.

This also answers the relabeling worry. Reliabilism, coherentism, correspondence theory, pragmatism, and social epistemology each isolate a real stabiliser. The reason to move beyond them lies in their relations: the history of correction among sources, the density of the causal network in which they operate, and the degree to which one route to success can compensate for or expose the failure of another. The epistemic fact is often carried by the pattern, not by a source considered alone.

The language-model case makes that structure visible because it pulls apart stabilisers that are ordinarily bundled together. A text-only model can inherit substantial textual residues of testimony and local patterns of inferential and discursive coherence. It can reproduce, recombine, and extend linguistic forms produced within human practices already partly accommodated to the world. That inherited participation still falls short of ordinary perceptual contact, calibrated measurement, intervention, or correction through action.

This is the profile explanation of hallucination and related failures. Humans also hallucinate, confabulate, misremember, and rely on testimony badly. The contrast lies in the standing correction profile. Human failures typically occur against a background in which perception, action, memory, testimony, and social correction remain available as mutually correcting routes. For a text-only model, several of those routes are absent or attenuated as part of the system's normal mode of operation. The same surface error can therefore have a different diagnostic significance.

The conclusion is comparative rather than classificatory. Retrieval, multimodal input, tool use, expert review, experimental embedding, and action-guided feedback alter the stabiliser profile. In some domains they may add genuinely new routes of answerability; in others they may import fresh text, automate a fragile proxy, or introduce new failure modes. The right question asks which stabilisers a given system participates in, how tightly those stabilisers are coupled, and what failures that

profile predicts under perturbation.

The account is deliberately empirical. If systems acquire reliable perceptual, instrumental, inter-ventive, or socially corrected routes to the world, their truth-tracking profile should change. If those changes do not weaken or alter the predicted failure signature, the profile has misdescribed the mechanisms. Conversely, where systems continue to depend mainly on inherited textual accommodation and local coherence, the same vulnerability should persist, even when surface fluency improves.

The broader lesson is not only about artificial systems. LLMs are useful pressure cases because they separate mechanisms that human epistemic life usually runs together. They show that truth-tracking is not exhausted by linguistic competence, internal coherence, or statistical sensitivity to past assertion. It depends on a wider ecology of correction. Once that ecology is represented as a profile, the interesting question becomes visible: what a system participates in, what it borrows, what it lacks, and where those absences should matter.

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